

EECE 5644 Assignment 02

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Practical Questions

1. What does the base plan cost?
 - a. Based on the intercept of the model, the base plan costs \$24.97/mo
2. How much does each individual add-on contribute?

Service	Coefficient	Price [\$ /mo.]
Phone Service	20.04	\$20.04
Multiple Lines	5.01	\$5.01
Online Security	5.04	\$5.04
Online Backup	4.98	\$4.98
Device Protection	5.01	\$5.01
Tech Support	5.02	\$5.02
Streaming TV	9.98	\$9.98
Streaming Movies	9.95	\$9.95
Fiber Optic Internet	24.95	\$24.95
Internet Service No	-25.05	-\$25.05

- a.
3. Which add-on is the most expensive, and which is the cheapest?
 - a. Fiber optic internet is the most expensive at \$24.95/mo
 - b. Online Backup is the cheapest at \$4.98/mo
 4. If a customer adds fiber internet plus both streaming services, how much should their bill rise?
 - a. Their bill would be expected to rise by \$44.88
 5. Given a specific bundle, what monthly charge should we expect?
 - a. A customer who wants to subscribe to phone service, streaming TV, and streaming movies would have an expected bill of \$64.93
 6. Is knowing which services a customer has meaningfully better than just knowing how many they have, for predicting the bill?
 - a. Yes, based on the r^2 value of each model, knowing only the number of services can explain ~70% of the variation in a customer's bill, while knowing the actual services they subscribe to can explain 99.8% of the variation in a customer's bill.
 7. How accurate are our predictions on customers the model hasn't seen? Within a few dollars, or off by more?
 - a. The MAE from applying the model to the reserved test data was 0.7886, meaning it is off by 79¢ on average for unseen data
 8. Do the model's learned prices match telco's actual price list, and are any coefficients surprising (negative, or larger/smaller than expected)
 - a. The model's coefficients appear to be within 5¢ of Telco's actual prices, there is one negative price, predicted for Internet Service No, which likely indicates that Telco offers a discount for customers who do not purchase internet service.